

Applied Zoology in Assam: Biodiversity, Economy, and Conservation

Assam and the broader North-East India region represent one of the most biologically rich and ecologically complex zones on the planet. Nestled within the Indo-Burma biodiversity hotspot, this region harbors extraordinary animal diversity — from the floodplain wetlands of the Brahmaputra and Barak river basins to the dense tropical forests of the Eastern Himalayas. Applied Zoology, as a discipline, bridges the gap between theoretical biological knowledge and real-world utilization of animal resources, making it especially relevant to a region where livelihoods, culture, and ecology are deeply intertwined.

This document explores six critical domains of Applied Zoology with a focused lens on Assam and North-East India: Fish & Fisheries, Biology of Insecta, Apiculture, Sericulture, Ornamental Fishery, and Wildlife Photography & Ecotourism. Each domain reflects a unique intersection of biodiversity conservation, economic opportunity, and cultural heritage. From the iconic Muga silk that has defined Assamese identity for centuries, to the thriving ornamental fish trade drawing on the region's riverine ecosystems, applied zoological practices are deeply embedded in the fabric of North-East Indian society.

Academic institutions across the region — including Gauhati University, Dibrugarh University, and Assam Don Bosco University — have taken the lead in developing curricula that integrate field-based learning with modern research methodologies. Under the framework of the National Education Policy (NEP-2020), these programs are being restructured to emphasize skill-based training, entrepreneurship, and sustainable resource management. The result is a new generation of zoologists, aquaculturists, and conservationists equipped to address both regional challenges and global ecological concerns.

APPLIED ZOOLOGY

NORTH-EAST INDIA

ASSAM

BIODIVERSITY

Fish & Fisheries

Assam's aquatic ecosystems are among the most productive and diverse in India. The state possesses approximately **100,815 hectares of floodplain wetlands** (locally known as *beels*) spread across the Brahmaputra and Barak river basins. These wetlands are not merely water bodies — they are dynamic ecological systems that support hundreds of indigenous fish species, sustain millions of livelihoods, and serve as critical buffers against flooding. The Brahmaputra alone is home to over 275 species of fish, including commercially important carps, catfishes, and air-breathing species uniquely adapted to the region's seasonal inundation cycles.

Among the most economically significant species are the **Indian Major Carps** — Rohu (*Labeo rohita*), Catla (*Catla catla*), and Mrigal (*Cirrhinus mrigala*) — which form the backbone of Assam's aquaculture industry. Indigenous catfishes like *Wallago attu* (the Wallago or freshwater shark) and air-breathers like *Clarias magur* (Magur) and *Anabas testudineus* (Kawai) are highly valued in local markets for their nutritional content and culinary significance. The seasonal floods that characterize Assam's hydrology create natural breeding grounds, making fishery management both an ecological and economic imperative.

Floodplain Wetlands	Indigenous Species	Livelihood Support
100,815 hectares of beels across Brahmaputra and Barak basins serve as critical fish habitats and natural breeding grounds.	Over 275 fish species including major carps, Wallago attu, Clarias magur, and endemic hill stream loaches.	Fisheries support over 2 million people in Assam, providing income, nutrition, and cultural identity to riverine communities.

Fishery enhancement strategies in Assam have evolved significantly over the past two decades. Government-led stocking programs, community-based wetland management, and the introduction of improved aquaculture techniques are transforming the sector. The Assam Fisheries Department, in collaboration with the Central Inland Fisheries Research Institute (CIFRI), has implemented programs to restock depleted beels with fingerlings of native species. These initiatives not only boost fish production but also help restore ecological balance in wetlands that have suffered from overfishing and habitat degradation. The integration of traditional ecological knowledge with modern fishery science represents a promising path toward sustainable utilization of Assam's aquatic resources.

Biology of Insecta

The North-East region of India, including Assam, is recognized globally as an **insect diversity hotspot**. The combination of tropical forests, high rainfall, varied topography, and relatively undisturbed ecosystems creates ideal conditions for an extraordinary range of insect life. Over **100 species of insects** have been documented within the Assam State Zoo complex alone – a figure that underscores the sheer richness of the region's entomological heritage. From the iridescent butterflies of the Kaziranga fringe forests to the diverse beetle communities of the Diphu hills, insects form the invisible backbone of North-East India's ecological networks.

Academic programs in Applied Zoology across Assam's universities place significant emphasis on insect biology. Courses cover insect taxonomy, physiology, ecology, and behavior, providing students with the foundational knowledge needed to understand insect roles in pollination, decomposition, pest dynamics, and food web interactions. The region's agricultural economy – heavily dependent on tea, rice, and horticulture – makes insect pest management a critical area of study. Understanding the life cycles, host preferences, and population dynamics of pest species like the tea mosquito bug (*Helopeltis theivora*) and the rice stem borer (*Scirpophaga incertulas*) is essential for developing integrated pest management (IPM) strategies that reduce chemical dependency.

Beyond pest management, insect biology in Assam also encompasses the study of beneficial insects – pollinators, parasitoids, and predators that support agricultural productivity and forest regeneration. Research on native bee species, silk moths, and medicinal insects like blister beetles is contributing to a deeper understanding of how insect diversity underpins regional sustainability. Field-based training programs in places like the Nameri National Park buffer zones and the Kaziranga-Karbi Anglong corridor provide students with hands-on experience in insect collection, identification, and ecological assessment.

Key Insect Groups in Assam

- Lepidoptera – Butterflies & Moths (including the Golden Birdwing)
- Coleoptera – Beetles (high diversity in forest litter)
- Hymenoptera – Bees, Wasps & Ants (critical pollinators)
- Odonata – Dragonflies & Damselflies (wetland indicators)
- Hemiptera – True Bugs (including agricultural pests)
- Diptera – Flies & Mosquitoes (disease vectors)

 The Eastern Himalayan region hosts over 50% of India's butterfly species, many of which are endemic to Assam and adjoining states.

Apiculture

Beekeeping, or apiculture, has emerged as a **key component of biological entrepreneurship** in Assam and the broader North-East region. The area's rich floral diversity — ranging from the mustard fields of the Brahmaputra Valley to the forest canopies of the Eastern Himalayas — provides year-round forage for honey bees, making it one of the most promising regions for apiculture in India. The dominant honey bee species in the region is the Asian honey bee (*Apis cerana indica*), though the rock bee (*Apis dorsata*) and the little bee (*Apis florea*) are also found in forested areas. The European honey bee (*Apis mellifera*) has been introduced in some areas for commercial honey production.

Apiculture is now formally integrated into agricultural and zoological curricula across Assam's universities and vocational training centers. The focus extends beyond honey production to encompass the broader ecological and economic roles of bees as pollinators. Studies have shown that bee pollination can increase crop yields by 20–30% in key Assamese crops like mustard, litchi, and citrus — making apiculture a critical component of agricultural productivity. Training programs emphasize modern hive management, honey extraction techniques, quality control, and market linkage, equipping students and rural entrepreneurs with the skills needed to build sustainable beekeeping enterprises.



Hive Management & Training

Modern apiculture training covers hive construction, colony management, disease control, and seasonal migration of hives to follow floral cycles across Assam's diverse landscapes.



Honey Production & Value Addition

Assam's forest honey, particularly from the Karbi Anglong and Dima Hasao districts, commands premium prices. Value-added products like beeswax, propolis, and royal jelly are gaining market traction.



Pollination Services

Beyond honey, bees provide critical pollination services to Assam's agricultural sector, enhancing yields in mustard, litchi, guava, and vegetable crops — a service increasingly recognized in economic terms.

The Assam State Apiculture Mission, supported by the National Bee Board, has been instrumental in promoting beekeeping among tribal communities and women's self-help groups. By harnessing the potential of natural floral resources for sustainable income generation, apiculture is proving to be one of the most accessible and ecologically sound livelihood options in rural Assam.

Sericulture

Sericulture is far more than an industry in Assam — it is a **cornerstone of cultural and economic identity**. The state is the only place in the world where all four commercially important silk types are produced: **Muga, Eri, Pat (Mulberry), and Tasar**. Of these, Muga silk — known for its natural golden-yellow luster and exceptional durability — holds a Geographical Indication (GI) tag and is synonymous with Assamese heritage. The *Directorate of Sericulture and Weaving, Assam*, manages a vast network of silkworm seed farms, mulberry plantations, and training centers that sustain this ancient craft across the state.

The biology of sericulture centers on the domesticated silk moths and their host plants. Muga silk is produced by the semi-domesticated silkworm *Antheraea assamensis*, which feeds exclusively on the leaves of *Persea bombycina* (Som) and *Litsea monopetala* (Sualu) — trees native to the Brahmaputra Valley. Eri silk, produced by *Samia ricini*, is known as "Ahimsa silk" because the pupa is not killed during extraction, making it a preferred choice for ethical textile consumers. Pat silk, from the mulberry silkworm *Bombyx mori*, is the most widely cultivated and forms the basis of Assam's commercial sericulture output.

 Muga Silk Golden-yellow, GI-tagged, produced by <i>Antheraea assamensis</i>. Exclusive to Assam. Used in traditional Mekhela Chador.	 Eri Silk "Ahimsa Silk" — pupa not killed. Produced by <i>Samia ricini</i>. Warm, wool-like texture. Popular in winter garments.
 Pat (Mulberry) Silk Produced by <i>Bombyx mori</i>. Most commercially cultivated. Bright white, fine texture. Basis of Assam's silk export trade.	 Tasar Silk Produced by <i>Antheraea mylitta</i>. Coarser texture, brownish color. Cultivated in forest-fringe areas of Assam and neighbouring states.

Academic programs in Applied Zoology at Gauhati University and other institutions include dedicated modules on sericulture biology, covering silkworm genetics, disease management (particularly grasserie and flacherie diseases), host plant cultivation, and reel technology. The integration of sericulture into the NEP-2020 skill-based curriculum is preparing a new generation of sericulturists who can combine traditional knowledge with modern biotechnology to improve silk quality and yield. The Assam Silk Mark initiative, which authenticates genuine Muga and Eri products, is also creating new market opportunities and protecting the interests of traditional weavers and rearers across the state.

Ornamental Fishery

Ornamental fishery is recognized as a **specialized and emerging area of study** in regional universities, with Assam Don Bosco University being among the pioneers in offering dedicated coursework in this domain. The North-East's riverine and wetland ecosystems harbor an extraordinary diversity of colorful, high-value fish species that are in demand both in domestic and international aquarium trade. Species like the Dwarf Loach (*Botia sidthimunki*), the Red-line Torpedo Barb (*Sahyadria denisonii*), and various species of Danio and Rasbora sourced from Assam's streams and beels are prized by aquarium enthusiasts worldwide for their vibrant coloration and unique behaviors.

The ornamental fishery sector encompasses three main areas of academic and professional focus: **trade and economics, fish pathology, and aquarium management**. Trade studies examine the supply chain from wild collection and captive breeding to export markets, with particular attention to sustainable harvesting practices that prevent overexploitation of wild populations. Fish pathology addresses the diseases and health management of ornamental species – a critical skill given the high mortality rates associated with poor water quality and stress during transport. Aquarium management covers tank design, water chemistry, breeding techniques, and species compatibility, preparing students for roles in both commercial and public aquarium settings.

Assam's ornamental fishery potential is closely linked to its wetland conservation agenda. Many of the most sought-after species inhabit the same beels and streams that are critical for food fisheries and biodiversity. Responsible ornamental fishery, therefore, requires careful regulation and community-based management to ensure that commercial interests do not compromise ecological integrity. The development of captive breeding protocols for high-value species is seen as a key strategy to reduce pressure on wild populations while creating sustainable livelihood opportunities for local communities.



275+

Fish Species

Documented in Brahmaputra basin,
many with ornamental potential

₹500Cr

Market Potential

Estimated domestic ornamental fish
market value in India

12

Export Species

North-East species currently in
international aquarium trade

Wildlife Photography & Ecotourism

Wildlife photography and ecotourism represent the convergence of **conservation science, visual storytelling, and sustainable economic development** in Assam. The state's iconic protected areas — Kaziranga National Park (a UNESCO World Heritage Site), Manas National Park, Nameri National Park, and the Dibru-Saikhowa Biosphere Reserve — are home to some of the world's most charismatic and endangered species, including the Greater One-horned Rhinoceros, the Wild Water Buffalo, the Bengal Florican, and the elusive Bengal Tiger. These landscapes provide unparalleled opportunities for nature-based tourism that, when managed responsibly, can generate significant revenue for conservation and local communities.

Institutions like the **Assam State Zoo and Botanical Garden** in Guwahati serve as critical hubs for ex-situ conservation and public environmental awareness. The Zoo houses over 895 animals representing 130 species, including several endangered and endemic species of North-East India. Beyond its conservation role, the Zoo functions as an educational platform where students of Applied Zoology gain practical experience in animal husbandry, veterinary care, and behavioral observation. The Botanical Garden complements this by preserving rare plant species and serving as a living laboratory for ecological studies.



Wildlife Photography

Training in ethical wildlife photography emphasizes minimal disturbance, species identification, and documentation of rare endemic species. Students learn to use photography as a tool for conservation advocacy and scientific record-keeping.



Ecotourism Management

Ecotourism programs cover guide training, interpretation skills, community engagement, and the economics of nature-based tourism. Emphasis is placed on ensuring that tourism revenue benefits local communities and funds conservation activities.



Conservation Education

Zoos and botanical gardens serve as platforms for public outreach, school programs, and citizen science initiatives. Documenting rare species through photography creates visual records that support research and policy advocacy.



Endemic Species Documentation

North-East India's endemic species — from the Pygmy Hog to the White-winged Wood Duck — require systematic documentation. Photography and citizen science platforms are increasingly used to monitor populations and track range changes.

Responsible nature-tourism in Assam is guided by principles that prioritize ecological integrity over commercial gain. The Assam Tourism Development Corporation, in partnership with forest departments and local communities, has developed community-based ecotourism models in areas like the Kaziranga buffer zones and the Brahmaputra river islands. These models demonstrate that wildlife photography and ecotourism, when integrated into applied zoology curricula, can produce professionals who are equally skilled in conservation biology and sustainable tourism management — a combination that is increasingly valued in the global environmental sector.

Academic Context: Building Regional Expertise

The academic infrastructure for Applied Zoology in Assam has grown substantially over the past three decades, driven by the recognition that the region's extraordinary biodiversity requires locally trained experts who understand both the science and the socio-cultural context of the North-East. **Gauhati University**, established in 1948, remains the flagship institution for zoological research and education in the region, offering undergraduate, postgraduate, and doctoral programs in Zoology with specializations in aquaculture, entomology, and wildlife biology. **Dibrugarh University**, located in Upper Assam, has developed strong research programs in fish biology and wetland ecology, leveraging its proximity to the Brahmaputra's dynamic floodplain ecosystems.

The **National Education Policy (NEP-2020)** has catalyzed significant curriculum reforms across Assam's universities. The policy's emphasis on multidisciplinary education, skill-based learning, and vocational integration has led to the introduction of new courses in aquaculture technology, sericulture management, apiculture entrepreneurship, and wildlife tourism. These courses are designed to bridge the gap between academic knowledge and industry requirements, ensuring that graduates are employment-ready from the moment they leave university. The introduction of Choice-Based Credit Systems (CBCS) and the flexibility to combine Zoology with courses in Environmental Science, Biotechnology, and Business Management reflects a holistic approach to building regional expertise.

Key Academic Institutions

- **Gauhati University** — Core research in zoology, aquaculture, and entomology
- **Dibrugarh University** — Fish biology, wetland ecology, and biodiversity studies
- **Assam Don Bosco University** — Ornamental fishery, biotechnology, and applied sciences
- **Assam Agricultural University** — Fisheries science, sericulture, and apiculture
- **North-Eastern Hill University (NEHU)** — Regional biodiversity and insect ecology

NEP-2020 Curriculum Highlights

- Skill-based courses in aquaculture and sericulture integrated into B.Sc. Zoology
- Internship and field training mandates with forest departments and fisheries
- Entrepreneurship modules for apiculture, ornamental fishery, and ecotourism
- Multidisciplinary options combining Zoology with Environmental Science and Biotechnology
- Research project requirements linked to regional conservation challenges

Beyond formal university education, a network of government training institutes, Krishi Vigyan Kendras (Farm Science Centers), and NGO-led programs provides vocational training in applied zoology disciplines to rural communities across Assam. These programs are particularly important for reaching tribal populations and women's self-help groups who may not have access to formal higher education but possess traditional ecological knowledge that is invaluable for sustainable resource management. The synergy between formal academic institutions and community-based training programs is creating a layered ecosystem of expertise that spans from laboratory research to field-level implementation.

Research and Conservation Thrust

Research and conservation form the backbone of Assam's applied zoology agenda. The region's unique biological heritage — shaped by its position at the confluence of the Indo-Malayan, Indo-Chinese, and Himalayan biogeographic zones — demands rigorous scientific investigation and proactive management. Major research centers across Assam are engaged in projects that span aquaculture genetics, wetland ecology, insect biodiversity, and wildlife conservation, often in collaboration with national and international institutions.

Bioinformatics Infrastructure Facility (BIF)

Located within Gauhati University, the BIF supports computational biology research on aquaculture species, silkworm genomics, and biodiversity informatics. The facility provides critical infrastructure for DNA barcoding, phylogenetic analysis, and database management of North-East India's animal diversity — tools that are essential for both conservation planning and commercial biotechnology applications.

National Bureau of Fish Genetic Resources (NBFGR)

Collaborative efforts with the NBFGR are vital to managing the North-East's unique fish genetic heritage. The Bureau's regional station works on characterizing indigenous fish germplasm, developing conservation strategies for threatened species, and supporting selective breeding programs that improve aquaculture productivity without compromising genetic diversity. This work is particularly important for species like the *Tor putitora* (Golden Mahseer) and various endemic catfishes that face habitat fragmentation and overfishing pressures.

Wetland Conservation & Management

Assam's beels are under threat from encroachment, pollution, and climate change. Research programs led by the Wetlands International South Asia office and the Assam Remote Sensing Application Centre are mapping wetland extent, assessing ecological health, and developing community-based management plans. These efforts are directly linked to fishery productivity, flood regulation, and biodiversity conservation — demonstrating the interconnected nature of applied zoology research in the region.

Wildlife Research & Monitoring

The Wildlife Institute of India (WII) and the Assam Forest Department collaborate on long-term monitoring of flagship species in Kaziranga, Manas, and Nameri. Camera trapping, radio telemetry, and population surveys provide the data needed for evidence-based conservation management. Research on human-wildlife conflict, habitat connectivity, and climate change impacts is informing policy decisions that affect millions of people and thousands of square kilometers of protected habitat.

The integration of traditional ecological knowledge with modern scientific methods is a defining feature of conservation research in Assam. Indigenous communities in the Karbi Anglong, Dima Hasao, and Bodoland regions possess generations of knowledge about local species, seasonal cycles, and sustainable harvesting practices. Research programs that engage these communities as partners — rather than subjects — are producing more culturally appropriate and ecologically effective conservation outcomes. This participatory approach to research is increasingly recognized as a model for biodiversity management in the Global South.

Conclusion: A Future Built on Sustainable Utilization

The six domains explored in this document – Fish & Fisheries, Biology of Insecta, Apiculture, Sericulture, Ornamental Fishery, and Wildlife Photography & Ecotourism – collectively represent a comprehensive vision for how Applied Zoology can serve as a driver of sustainable development in Assam and North-East India. Each domain demonstrates that the region's extraordinary animal diversity is not merely a conservation concern but an economic asset that, when managed responsibly, can generate livelihoods, preserve cultural heritage, and fund the conservation activities that protect biodiversity for future generations.

Biodiversity as Asset Assam's 275+ fish species, 100+ insect species in zoo complexes, four silk types, and iconic wildlife represent irreplaceable biological capital that underpins regional economies.	Academic Innovation NEP-2020 curriculum reforms, skill-based courses, and university-industry partnerships are producing a new generation of applied zoologists equipped for the regional job market.
Conservation Integration Research centers, bioinformatics facilities, and collaborative networks with national institutions ensure that utilization is always guided by sound science and long-term ecological thinking.	Cultural Continuity From Muga silk weaving to traditional fishery practices, applied zoology in Assam honors and strengthens the cultural traditions that have sustained communities for millennia.

Looking ahead, the integration of applied zoology into academic, professional, and policy sectors provides the necessary tools for economic development that does not come at the expense of ecological integrity. By leveraging the rich animal diversity of the Brahmaputra Valley, the Eastern Himalayas, and the North-East's wetland systems – combined with innovative management practices, modern biotechnology, and community-centered approaches – Assam is uniquely positioned to lead the North-East region in sustainable biological resource utilization. The future of applied zoology in Assam is not simply about studying animals; it is about building a society where human well-being and ecological health are mutually reinforcing, where traditional knowledge and modern science converge, and where the golden threads of Muga silk and the shimmering scales of ornamental fish tell a story of a region that has found its path to prosperity through the wise and respectful use of its natural heritage.

-  Assam's applied zoology sector represents a model for sustainable biological resource utilization – one that balances economic development, cultural preservation, and ecological conservation in a way that few regions in the world can match.